Report — Comparison of Optimization Methods for a PDE-Constrained Optimal Control Problem

Fachseminar

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1 Continuous Problem

We consider the PDE-constrained optimal control problem on a bounded Lipschitz domain $\Omega = (0,1)^2 \subset \mathbb{R}^2$ with a measurable control region $\omega \subset \Omega$. The state $y\colon \Omega \to \mathbb{R}$ and the control $u\colon \omega \to \mathbb{R}$ are coupled through the Poisson equation with homogeneous Dirichlet boundary conditions:

$$-\Delta y = \chi_\omega\, u \qquad \text{in } \Omega, \tag{1}$$

$$y = 0 \qquad \text{on } \partial\Omega. \tag{2}$$

Given a desired state $y_d \in L^2(\Omega)$ and a Tikhonov parameter $\beta > 0$, we minimize the quadratic objective

$$F(y,u) = \tfrac{1}{2}\, \|y - y_d\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\, \|u\|^2_{L^2(\omega)}\,. \tag{3}$$

We adopt the space setting

$$y \in H^1_0(\Omega), \qquad u \in L^2(\omega), \tag{4}$$

where the right-hand side $\chi_\omega u$ is interpreted as an element of $H^{-1}(\Omega)$ via the embedding $L^2(\omega) \hookrightarrow H^{-1}(\Omega)$.

Well-posedness of the state equation follows from Lax–Milgram: for each $u \in L^2(\omega)$ there exists a unique $y \in H^1_0(\Omega)$ such that the weak form below holds. Existence and uniqueness of an optimal control follow from strict convexity and coercivity of F.

2 Weak Formulation

Let $V := H_0^1(\Omega)$. The weak form of the state equation reads: find $y \in V$ such that

$$a(y, v) = \ell(v; u) \qquad \forall v \in V, \tag{5}$$

with the bilinear form and linear functional given by

$$a(y, v) := \int_\Omega \nabla y \cdot \nabla v \, dx, \qquad \ell(v; u) := \int_\omega u \, v \, dx. \tag{6}$$

The control-to-state map $S\colon L^2(\omega) \to V$ is defined by $y = Su$ as the unique solution of the above variational problem.

To derive first-order optimality conditions, we introduce the adjoint state $p \in V$ defined by

$$a(w, p) = \int_\Omega (y - y_d) \, w \, dx \qquad \forall w \in V. \tag{7}$$

The reduced formulation in the control (see Sections 5 and 6) yields the optimality condition in $L^2(\omega)$: $\nabla j(u) = 0$ with an explicit expression for the gradient via y and p.

3 Finite Element Discretization

We consider a conforming triangulation $\mathcal{T}_h$ of Ω and the continuous, piecewise linear finite element space

$$V_h := \{v_h \in C^0(\overline{\Omega}) : v_h|_K \text{ is affine } \forall K \in \mathcal{T}_h,\ v_h|_{\partial\Omega} = 0\} \subset V. \tag{8}$$

For the control, we use a compatible discrete space $U_h \subset L^2(\omega)$. In the simplest case, we take the V_h basis restricted to the control region and do not enforce boundary conditions on U_h:

$$U_h := \operatorname{span}\{\phi_i|_\omega : \phi_i \text{ nodal basis of } V_h\}, \tag{9}$$

which is standard when the control acts as a distributed source over ω.

The discrete state $y_h \in V_h$ for a given $u_h \in U_h$ solves

$$a(y_h, v_h) = \ell(v_h; u_h) \qquad \forall v_h \in V_h. \tag{10}$$

Homogeneous Dirichlet conditions are imposed on V_h (e.g., by modifying the stiffness matrix rows/columns associated with boundary nodes), while no essential conditions are applied to U_h.

4 Discrete Operators

Let $\{\phi_i\}_{i=1}^{n_V}$ be the nodal basis of V_h and $\{\psi_j\}_{j=1}^{n_U}$ a basis of U_h. We define the usual FE matrices:

$$A_{ij} := \int_\Omega \nabla\phi_i \cdot \nabla\phi_j \, dx \qquad \text{(stiffness on } \Omega), \tag{11}$$

$$M_{ij} := \int_\Omega \phi_i \, \phi_j \, dx \qquad \text{(mass on } \Omega), \tag{12}$$

$$(M_U)_{jk} := \int_\omega \psi_j \, \psi_k \, dx \qquad \text{(mass on } \omega), \tag{13}$$

$$B_{ij} := \int_\omega \psi_j \, \phi_i \, dx \qquad \text{(control coupling)}. \tag{14}$$

With these definitions, the discrete state and adjoint equations read

$$Ay = Bu, \tag{15}$$
$$Ap = M(y - y_d), \tag{16}$$

where $y, p \in \mathbb{R}^{n_V}$ and $u \in \mathbb{R}^{n_U}$ are the coefficient vectors, and $y_d \in \mathbb{R}^{n_V}$ is the V_h projection of the data. Dirichlet boundary conditions are applied to A (and the corresponding right-hand sides) in the standard way; the matrices M_U and B are assembled using cell/subdomain tags so that integrations over ω are accurate.

5 Reduced Functional

Eliminating the state via $y = A^{-1}Bu$ yields the discrete reduced cost

$$j(u) := \tfrac{1}{2}(y - y_d)^\top M(y - y_d) + \tfrac{\beta}{2}\, u^\top M_U u, \qquad y = A^{-1}Bu. \tag{17}$$

Expanding in u gives the quadratic form

$$j(u) = \tfrac{1}{2}\, u^\top \underbrace{\left(B^\top A^{-1} M A^{-1} B + \beta M_U\right)}_{\widetilde{H}} u - u^\top B^\top A^{-1} M y_d + \tfrac{1}{2}\, y_d^\top M y_d, \tag{18}$$

which shows j is strictly convex for $\beta > 0$. The constant term does not affect optimization.

6 Gradient and Hessian

We equip the control space with the U-inner product

$$(a, b)_U := a^\top M_U b, \qquad \|a\|_U^2 = (a, a)_U\,. \tag{19}$$

For a given u, compute the state y and the adjoint p from

$$Ay = Bu, \qquad Ap = M(y - y_d). \tag{20}$$

Then the Riesz-represented gradient in the U-inner product is

$$\nabla j(u) = M_U^{-1}(B^\top p + \beta M_U u) = M_U^{-1} B^\top p + \beta u. \tag{21}$$

Equivalently, the U-duality gradient is $g_U(u) := B^\top p + \beta M_U u$. The (Riesz) Hessian operator is the SPD map

$$H = M_U^{-1}(B^\top A^{-1} M A^{-1} B + \beta M_U), \tag{22}$$

so that for any direction d the Hessian-vector product can be computed by

$$\text{solve } A\, y'[d] = B\, d, \tag{23}$$
$$\text{solve } A\, p'[d] = M\, y'[d], \tag{24}$$
$$\text{return } Hd = M_U^{-1}(B^\top p'[d] + \beta M_U d). \tag{25}$$

This structure underpins fixed-step gradient descent with step $1/L$, Barzilai–Borwein steps using the U-metric, and Nesterov acceleration with restarts.